

Operations Analytics

Optimise operations through analytics and modelling.

Programme level	Graduate, Postgraduate	Duration	12 weeks
Credits	3	Delivery	Classroom, Online, Hybrid
Subject area	Operations Analytics	Updated	2026-07-21

Course overview

An applied course on process analysis, capacity, forecasting, inventory, scheduling and optimisation for operational performance.

Intended audience: Operations, supply-chain, engineering management and business analytics students and professionals.

Prerequisites: Basic quantitative methods and spreadsheet modelling.

Learning outcomes

- Measure and diagnose operational performance.
- Develop demand and workload forecasts.
- Analyse capacity, queues, inventory and service levels.
- Formulate scheduling and optimisation decisions.
- Recommend operational improvements using analytical evidence.

Module structure

01	Operational performance and process data Flow, capacity, productivity and service measures.
02	Forecasting for operations Demand, workload and forecast evaluation.
03	Capacity, queues and service Congestion, resource decisions and service trade-offs.
04	Inventory and scheduling analytics Policies, priorities and resource allocation.
05	Optimisation and implementation Models, scenarios and operational adoption.

Assessment pattern

Assessment component	Weight
Operations modelling exercises	30%
Process improvement case	30%
Operations analytics project	40%

Delivery note

The module sequence, class activities and assessment details may be adapted for programme schedules, cohort experience and institutional requirements. The course remains focused on practical application, structured judgement and professional communication.